

Children’s emotional regulation

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Older children show stronger emotional regulation than younger children.

Introduction

Children’s ability to regulate emotions improves as they grow, but the pace and patterns of this development vary. This study examines how emotional regulation scores differ across age groups in a sample of children aged 5–12. We expected older children to regulate their emotions better than younger children.

Methods

Participants and Data

Data were collected from a sample of children aged 60-150 months, each of whom completed an emotional regulation assessment in an online survey alongside demographic data. The dataset included age in months and an emotional regulation score.

Data Preparation

Age in months was converted into years to make results easier to interpret. The dataset was then cleaned by removing missing values on the key variables.

```
# emotional_regulation is too long name. rename it.
df <- df %>%
  rename( # function rename
    emoreg = emotional_regulation)
# variable now called 'emoreg'

# mutate the months variable into age
df <- df %>%
  mutate(age = (months/12))
# age is the new variable which is months
# variable divided by 12

# Check for missing values in key variables
colSums(is.na(df[, c("age", "emoreg")]))
```

```
##      age emoreg
##      1       2
```

```
# There are one missing in age and 2 in emoreg
# Remove missing values ONLY for relevant variables

df_clean <- df %>%
  drop_na(age, emoreg)
```

Descriptive and Inferential Statistics

Descriptive statistics were computed for age and emotional regulation and summarised in a table and density plots. A simple linear regression tested whether age predicted emotional regulation. A scatterplot with a fitted regression line and 95% confidence interval was used to visualise the relationship, and a Q-Q plot was used to assess residual normality.

Descriptive Results

The dataset included 20 children aged 5–12 years (mean age = 8.25 years, SD = 2.38). Emotional regulation scores ranged from 3.5 to 12.0, with a mean of 8.44 (SD = 2.10).

Table 1: Summary Statistics for Age and Emotional Regulation

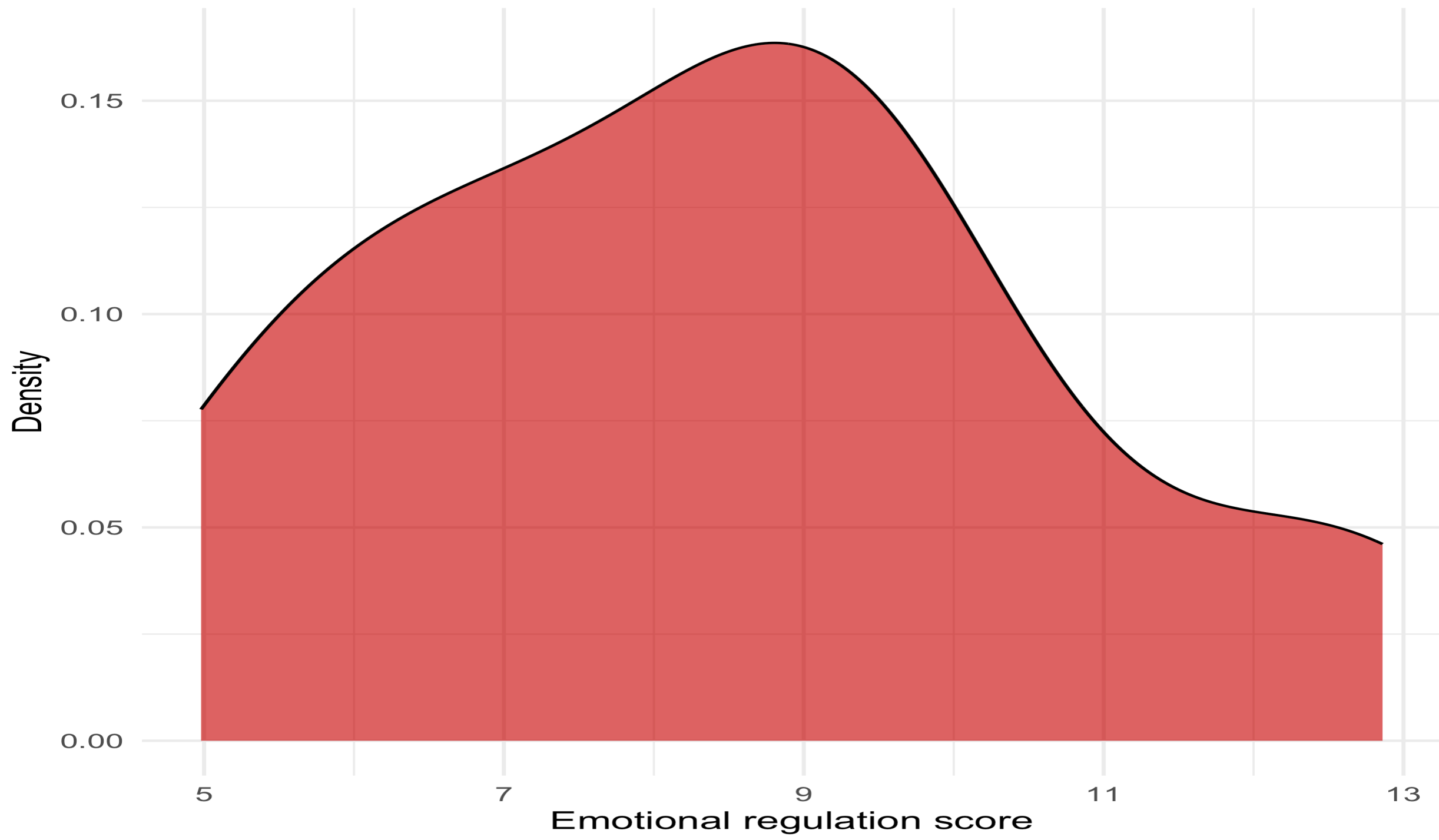
Mean_Age	SD_Age	Mean_ER	SD_ER
8.35	2.10534	8.441	2.270917

Density plots of the two variables show that age was approximately normally distributed, while emotional regulation scores were slightly positively skewed, with most children scoring between 6 and 10.

Figure 1. Density plot for Age



Figure 2. Density plot for emotional regulation



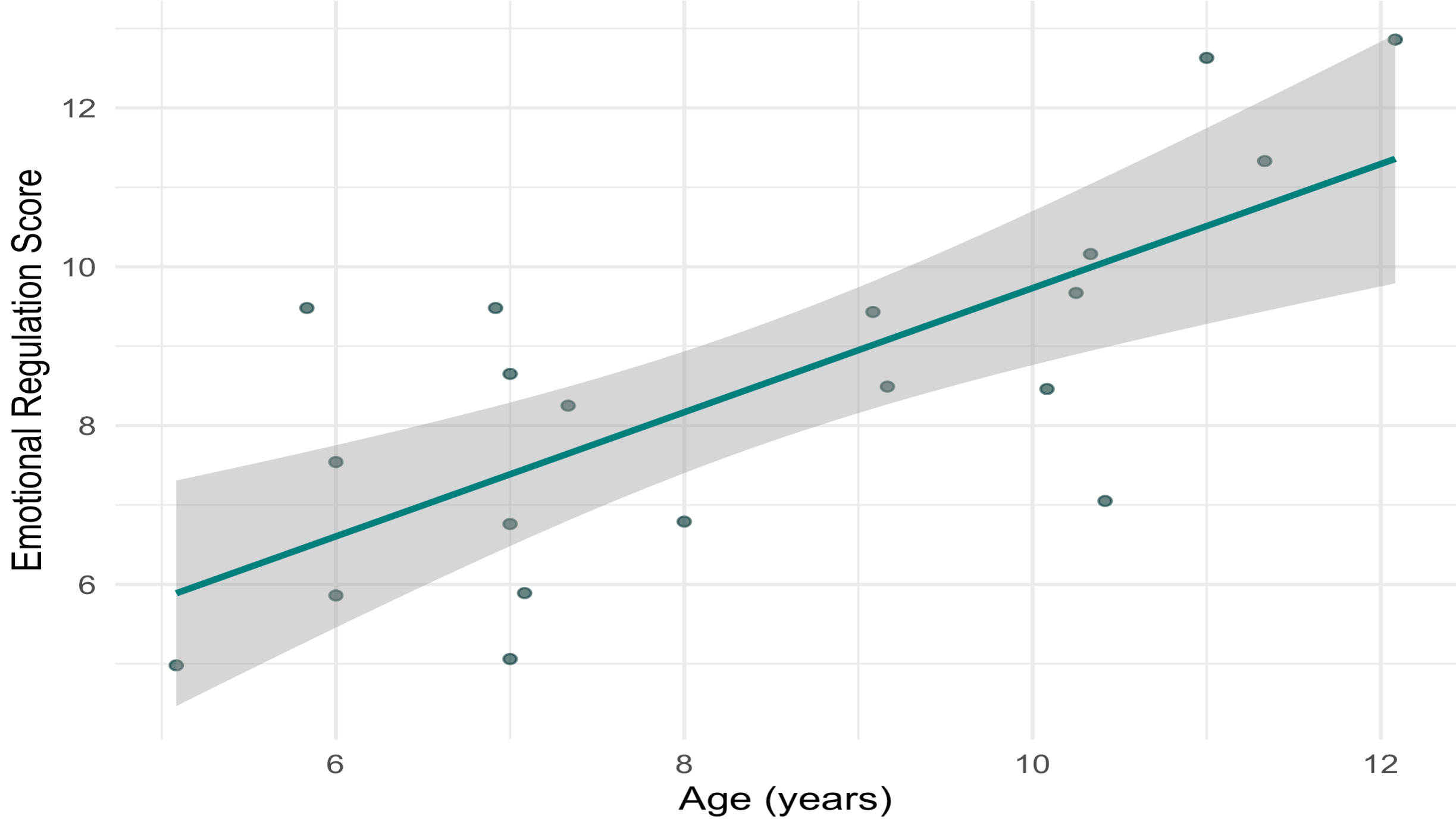
Regression Results

A simple linear regression was conducted to examine whether age predicted emotional regulation in children aged 5-12. The model was significant, $F(1, 18) = 28.71$, $p < .001$, and explained a substantial proportion of the variance in emotional regulation, $R^2 = 0.61$. Age was a significant positive predictor of emotional regulation, $B = 0.80$, $SE = 0.15$, $p < .001$, indicating that older children tended to have higher emotional regulation scores.

Table 2: Linear regression model results

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.915	1.507	1.271	0.22
age	0.782	0.175	4.461	0.00

Figure 3. Scatter plot for the relationship between age and emotional regulation



Residuals were checked for normality and Q-Q plots showed approximate normality.

Conclusion

This study demonstrates that emotional regulation improves with age in children aged 5-12. Older children showed stronger self-regulation than younger children, consistent with typical developmental trends. These findings highlight the importance of supporting emotional skills in early childhood to promote healthy development.